

Project Developing Phase

Preprocessing Steps and Business Questions with Visualizations Template

Date	03 July 2026
Team ID	
Project Name	Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau
Maximum Marks	

Preprocessing steps:

Step 1:

Import the UNESCO World Heritage Dataset:

The UNESCO World Heritage Sites dataset was collected in CSV format from a reliable public data source. The dataset contains information about World Heritage Sites, including site name, country, region, category, year of inscription, geographical coordinates, and conservation status. The dataset was imported into Tableau Desktop for further preprocessing and visualization.

Step 2:

Data Cleaning:

- **Remove duplicate records:**

The dataset was examined for duplicate entries to ensure that each UNESCO World Heritage Site appeared only once. Duplicate records, if found, were identified and removed to improve the accuracy and reliability of the analysis.

- **Handle missing values:**

The dataset was checked for missing or null values in important columns such as Site Name, Country, Category, Region, Year of Inscription, and Coordinates. Missing values were handled by replacing them with appropriate values where possible or excluding incomplete records that could affect the analysis.

- **Correct inconsistent values:**

The dataset was standardized by correcting inconsistent spellings, formatting differences, and naming conventions. Country names, heritage categories, and regional values were made consistent to ensure accurate grouping, filtering, and visualization throughout the project.

Step 3:

Data Transformation:

- **Rename columns:**
Column names were renamed to make them more meaningful and easier to understand. For example, lengthy or unclear column names were simplified while maintaining their original meaning, improving readability during dashboard development.
- **Change data types:**
Data types were verified and modified according to the nature of the data. Text fields such as Site Name and Country were assigned string data types, Year of Inscription was converted to a numeric/date type, and Latitude and Longitude were maintained as geographic fields for map visualization.
- **Create calculated fields:**
Calculated fields were created to support analytical visualizations and dashboard metrics. Calculations such as total heritage sites, percentage distribution of heritage categories, unique country count, and endangered site classification were used to generate meaningful insights and interactive visualizations in Tableau.

Step 4:

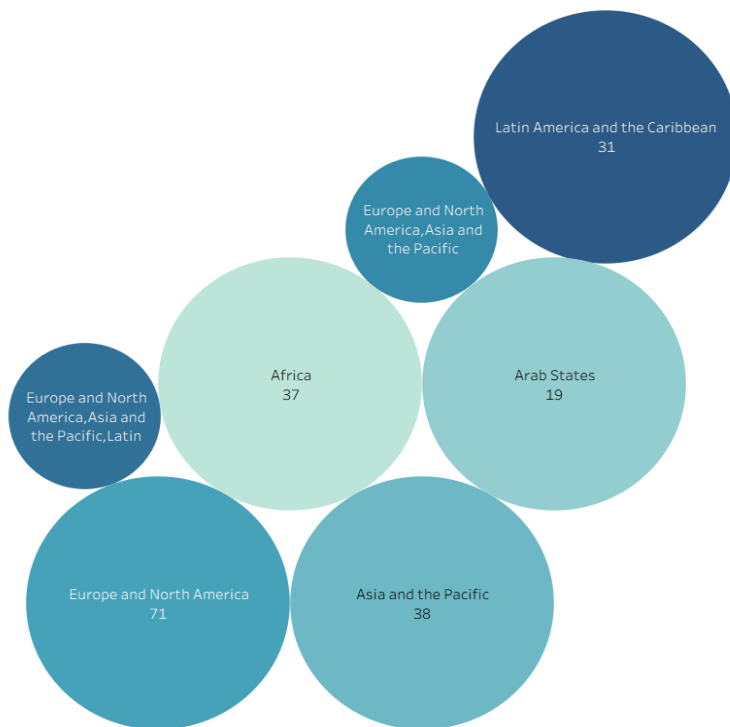
Load the Cleaned Dataset into Tableau:

- **Load the cleaned dataset into Tableau:**
After completing data cleaning and transformation, the processed dataset was loaded into Tableau Desktop. The cleaned data was then used to create interactive dashboards, charts, maps, KPIs, filters, and story points that provide comprehensive insights into UNESCO World Heritage Sites across different countries, regions, categories, and historical periods.

Business Questions with Visualizations:

1. Which UNESCO regions have the highest and lowest number of participating countries, and how are countries distributed across different regions?

Countries per Region

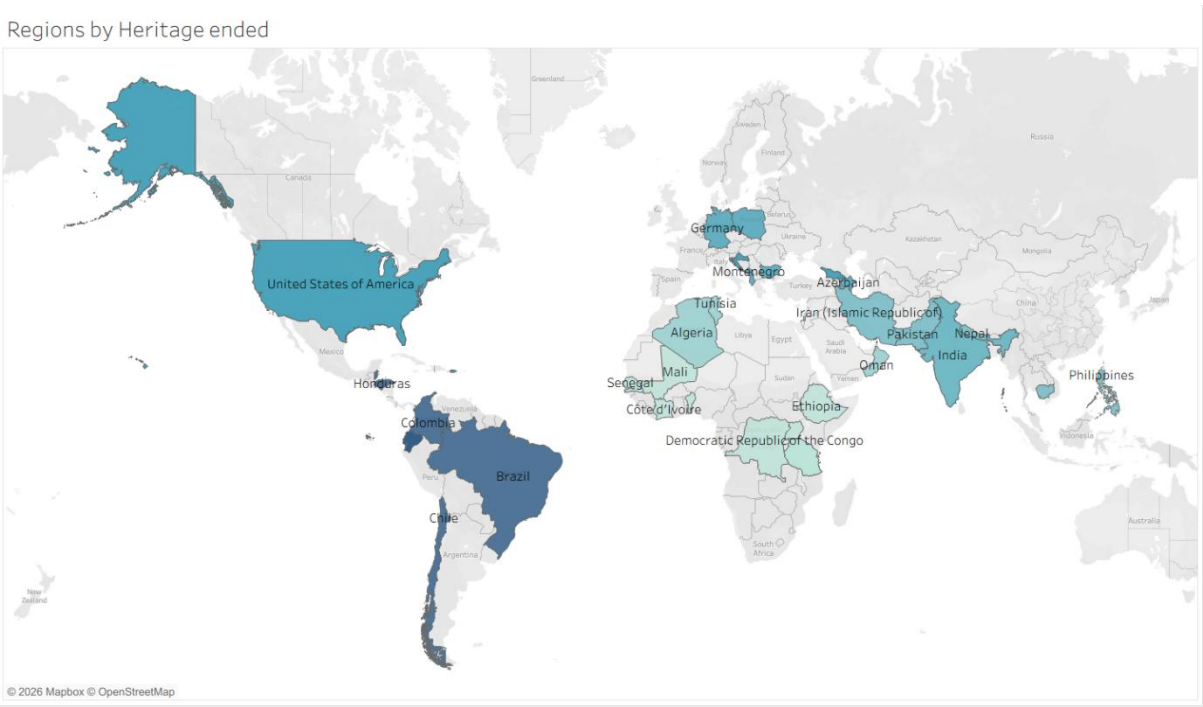


2. Which are the top 10 UNESCO regions with the largest total heritage site area, and how do they compare with one another?

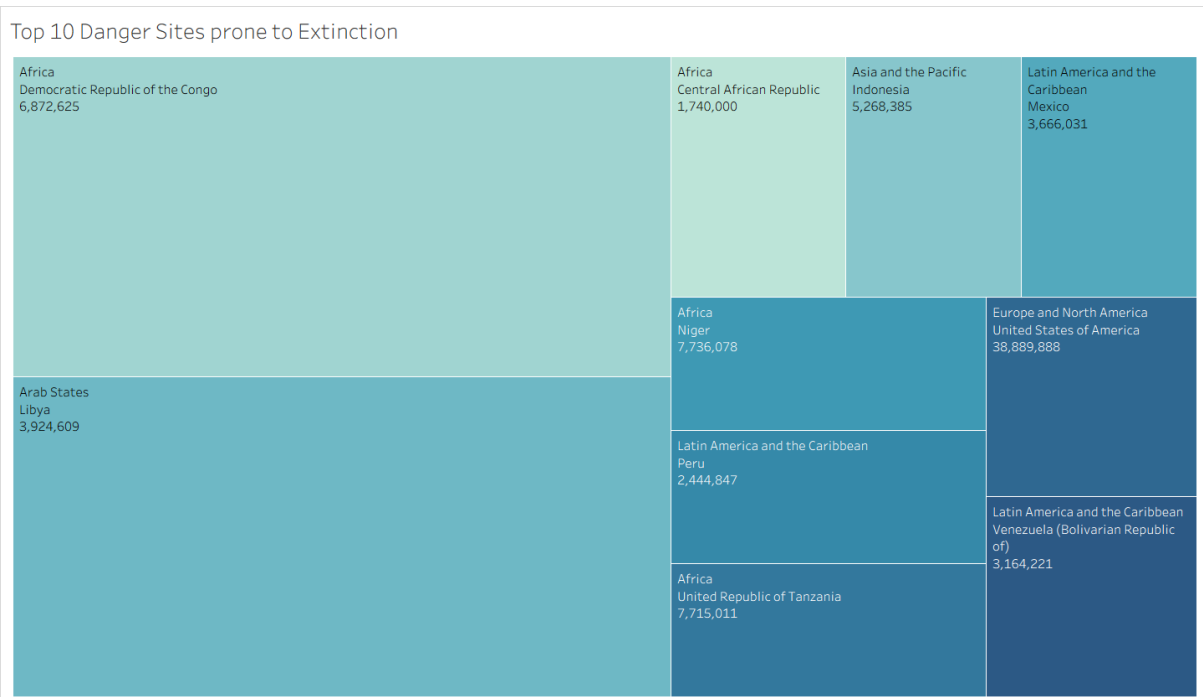
Top 10 regions by area



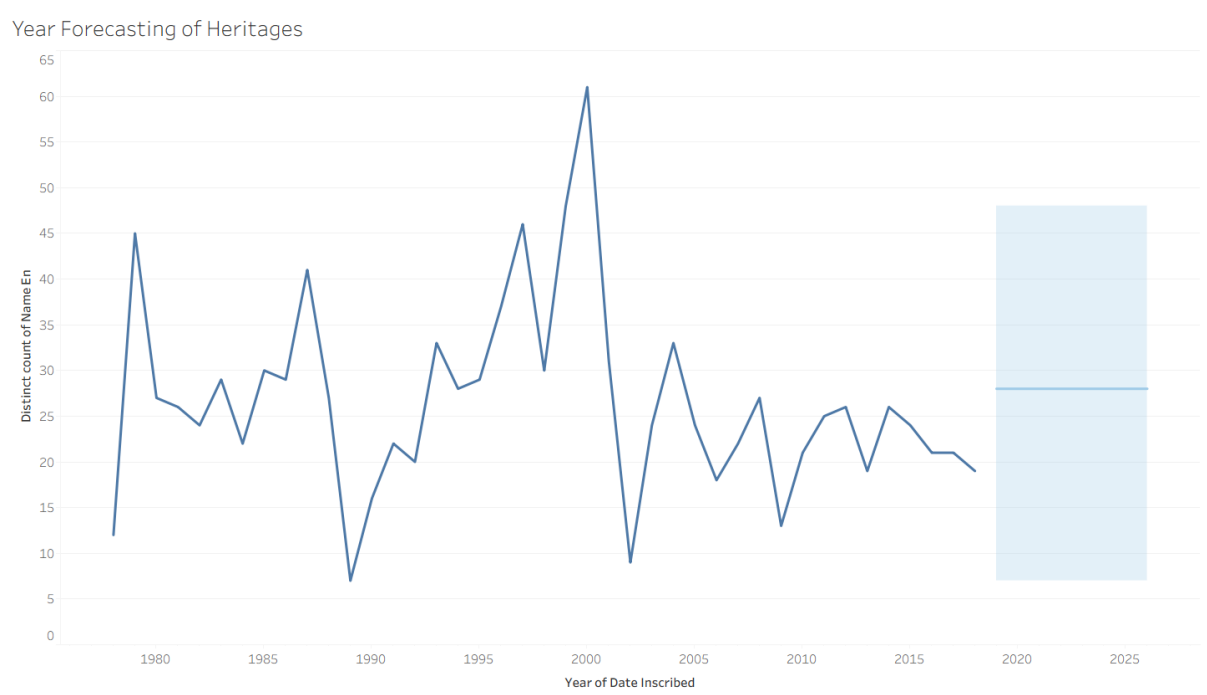
3. Which regions have the highest number of UNESCO World Heritage Sites classified as endangered?



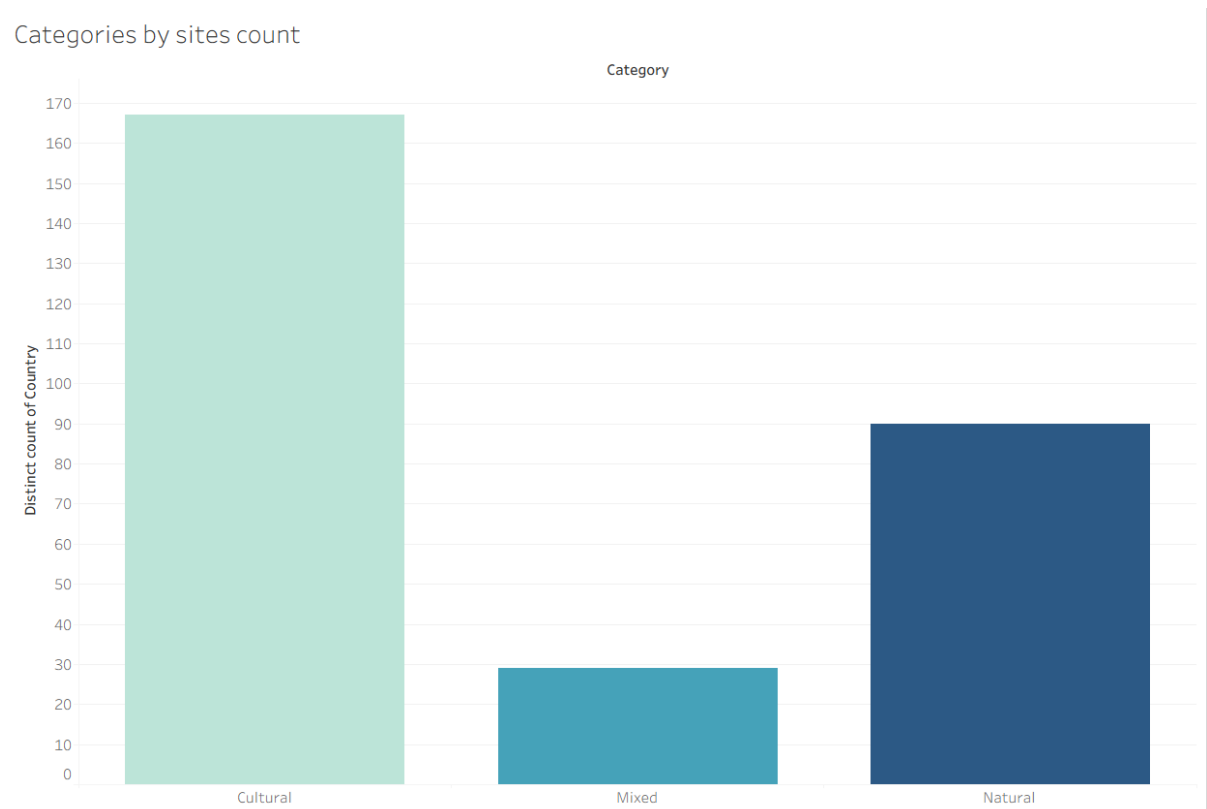
4. Which are the top 10 UNESCO World Heritage Sites that are most prone to extinction or currently classified as endangered?



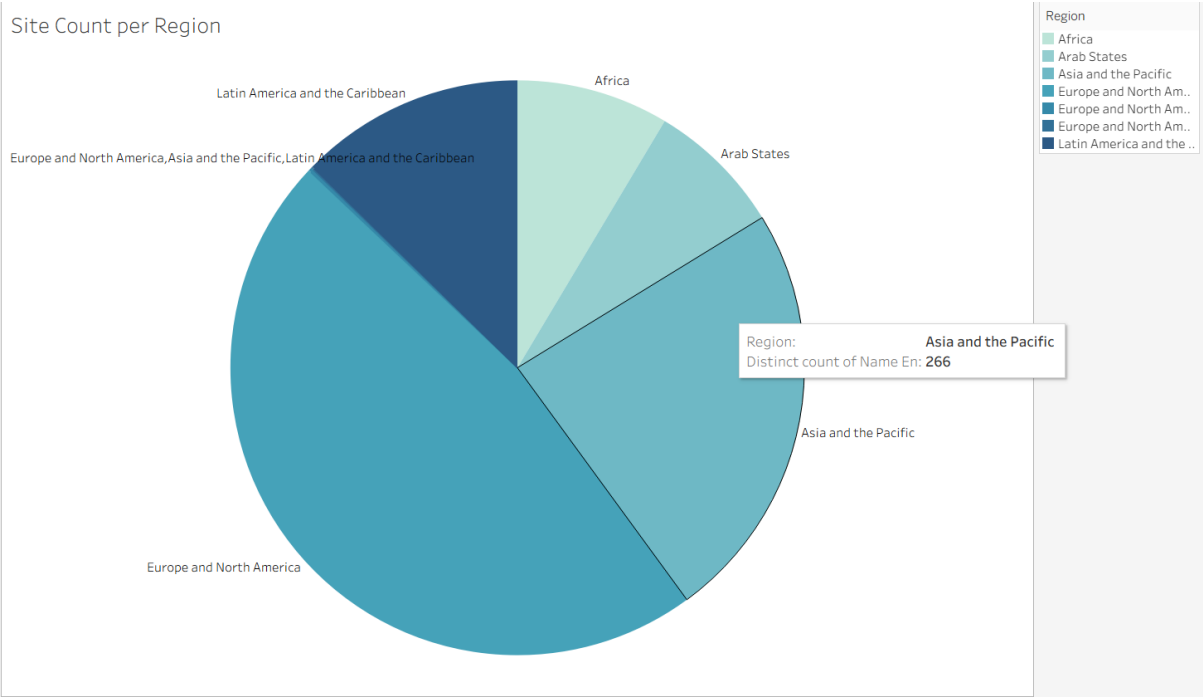
5. How is the number of UNESCO World Heritage Sites expected to grow in the coming years based on historical trends?



6. How are UNESCO World Heritage Sites distributed across different heritage categories based on their site count?



7. How are UNESCO World Heritage Sites distributed across different regions based on site count?



8. How does the area of UNESCO World Heritage Sites relate to their endangered status across different regions?

